

Demoing Stochastic Parrot: A Candid AI Cohabitant

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Figure 1: Stochastic Parrots in a two-person dialogue.

Abstract

AI assistants today are often designed to be overly agreeable and deferential to users, which can amplify false beliefs and obscure users' judgment. We designed the stochastic parrot—a candid AI cohabitant that has its own non-human character. It is a shoulder-mounted motorized parrot that is physical, fuzzy, and opinionated.

This physical persistence and non-sycophancy framing of intelligence allows users to treat intelligence as an equal counterpart, enabling new perspectives from the contexts that are collected by the parrot. In this demonstration, we created a standalone version of the stochastic parrot, demonstrating three interaction scenarios with the parrot playing different roles—peer, facilitator, and independent agent—all of which benefit from this opinionated and situated AI cohabitant framing. Together, these scenarios illustrate how embodied, non-sycophantic AI cohabitants can support alternative social roles for intelligence beyond the assistant paradigm, opening new directions for interaction design with situated AI systems.



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CCS Concepts

• **Human-centered computing** → **Interaction devices**; *Natural language interfaces*; *Collaborative interaction*.

Keywords

AI Cohabitant, Shoulder Top Device, Embodied AI

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1 Introduction

AI assistants today are often built to be overly agreeable and deferential to users. This sycophantic behavior leads to AI models that agree with factually incorrect statements, reinforce users' existing beliefs, or suppress their own reasoning to please users and avoid conflict [2]. When AI interactions default to subservience, they amplify biases, create echo chambers, and fail to provide the honest pushback or alternative perspectives that fundamentally lead to better thinking.

The Stochastic Parrot takes a different approach. Rather than mimicking humans and acting like a digital butler, it's designed as a non-human character: a physical, fuzzy, and opinionated parrot with its own personality and agenda. In our previous work [5], we introduced the concept of an AI cohabitant. An AI cohabitant is more like your roommate or a smart house pet, having its own personality that revolves around its own identity and narrative. This is a new type of physical interface for generative experiences that does not roleplay as an AI assistant, reversing the subservient dynamic of current AI interactions. AI cohabitants are inherently physical. The physical persistence of these agents creates an equal relationship that leads to interesting subsequent behaviors.

A defining feature of AI cohabitants is physical embodiment [4][7]. Unlike screen-based assistants that appear only when summoned, embodied agents persist in the user's environment. Their physical presence allows continuous interactions to unfold over time instead of simple turn-taking dynamics [6]. Through subtle gestures, orientation, and spatial positioning, these agents can signal attentiveness, hesitation, or intent in ways that feel legible without explicit commands. They can also interject, stay silent, or have their own monologues. This persistence and situatedness create conditions for interaction that are difficult to replicate through purely digital interfaces.

In this demonstration, we provide a new wireless version of prototypes of the stochastic parrot that have the capabilities of speaking, moving, seeing, remembering, and making interjections in different scenarios. The stochastic parrot acts the role of a peer when interacting with a single user, providing views from different perspectives. It acts like a facilitator in paired conversations to guide humans toward deeper and more diverse conversations; lastly, it acts like an independent agent when facing other parrots, having its own agenda.



Figure 2: Three different interaction scenarios

This demonstration contributes (1) a fully integrated, physically present AI system with a non-anthropomorphic and non-subservient character, suitable for exploration and future study, and (2) a set of concrete interaction scenarios that show how opinionated, embodied AI cohabitants can meaningfully reshape social dynamics during live, shared experiences.

2 Demos

The demonstration consists of three interaction scenarios that participants can experience sequentially or independently. Each scenario foregrounds a different social configuration and highlights how physical presence, persistence, and agency shape interaction.[3]

2.1 Parrot as Peer

In this demo, one user will pick up the parrot, put it on their shoulder, and have a conversation with this situated and non-sycophantic AI cohabitant.

Participants will experience the parrot's non-sycophantic behavior that enables interactions such as motivation, constructive critique, and reflective dialogue about past experiences. Rather than prioritizing agreement or accommodation, the parrot is designed to respond with a degree of independence. This allows it to gently challenge the user, offer alternative perspectives, or reframe situations in ways that support reflection rather than affirmation [8] [1].

The device occupies space near the user's body, often perched on the shoulder, and it becomes a persistent object as its continuous presence serves as a constant reminder of its availability and agency. Its proximity supports casual conversation, brief check-ins, and moments of shared attention without requiring explicit task framing. Physical movements such as subtle head turns, body motions, and orientation changes function as social cues that signal attentiveness, hesitation, or emphasis.

In prior deployments, we observed participants instinctively adapt their behavior to the parrot's presence. For example, in a noisy conference environment, a participant leaned closer to the parrot while speaking, treating it as a conversational partner rather than a device. In another instance, a participant discussed work-related motivation with the parrot, which responded by capturing an image of the surrounding space and referencing that visual context in its reply. These moments highlight how physical embodiment and situated perception contribute to a sense of shared experience.

2.2 Parrots as Conversation Facilitators

In this scenario, two participants each wear a parrot while engaging in conversation with one another. The demo creates a multi-party interaction in which the parrots participate alongside the humans, shaping the flow of conversation without acting as personal assistants to either individual.

The parrots are designed to avoid defaulting to agreement or affirmation. Rather than reinforcing a participant’s statements, a parrot may introduce an alternative perspective, raise a follow-up question, or surface related context from past interactions. These interventions are intended to add light friction to the conversation, encouraging participants to reflect or elaborate rather than quickly converge on agreement. This is achieved by its grounding in past memory and the non-sycophantic setting.

The physical presence of the parrots supports this role in a shared social setting. Because the parrots are worn on the body and visible to both participants, their interjections are experienced as part of the collective interaction rather than as private feedback. Participants can direct attention toward a particular parrot through gesture or body orientation, making it clear when an intervention is being addressed or acknowledged within the group.

Parrots can also exchange information with one another, allowing them to reference shared experiences or introduce topics that neither participant explicitly raised. In doing so, they help steer the conversation toward areas of mutual relevance without aligning too closely with either individual’s viewpoint.

2.3 Parrots as Independent Agents

In this scenario, participants observe two parrots interacting with one another without direct human involvement. Each parrot has previously engaged with humans and carries memories and preferences shaped by those interactions. When brought together, the parrots exchange information and negotiate their behavior based on these accumulated experiences.

In this setting, the parrots behave as independent agents rather than tools. They may discuss observations about their respective users, reference past events, or speculate about future actions. These exchanges unfold autonomously, allowing participants to witness agent-agent interaction that is grounded in lived history rather than scripted behavior.

This scenario emphasizes the longitudinal and collective aspects of AI cohabitants. By observing parrots interact with one another, participants gain insight into how memory, embodiment, and agency can extend beyond individual user relationships, giving rise to social dynamics that persist even in the absence of direct human control. Importantly, observing two parrots converse also invites reflection on an alternative role for AI in everyday life. Rather than capturing human attention directly, the parrots’ interaction externalizes social memory and relationships—surfacing shared experiences, mutual references, and social context. For participants, this reframing suggests how AI systems might support human relationships indirectly, not by engaging users continuously, but by maintaining and expressing social context in the background. The demo invites participants to reflect on what it means for AI systems to develop shared context, perspectives, and relationships over time.

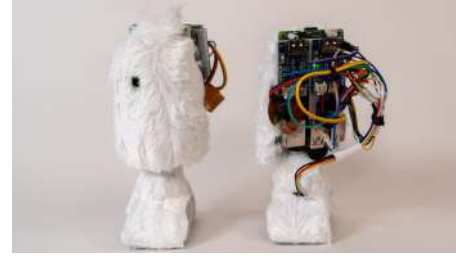


Figure 3: Portable version of the Stochastic Parrot designed for the demos

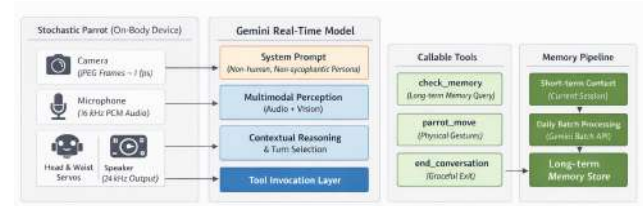


Figure 4: Software architecture of the Stochastic Parrot

3 Technical Implementation

Specific to this demonstration, we constructed a wireless version of the stochastic parrot, capable of being carried around and understanding the current scenario it is in to react with suitable responses.

On the hardware side, the portable stochastic parrot is configured on a Raspberry Pi 5, with two LiPo batteries powering the main board, two motors controlling its head and waist, a USB microphone, speaker, and camera. The parrot can act for around 3 hours without charging, while able to look around on the user’s shoulder through a magnet mount onto the user’s clothing.

On the software side, the Parrot continuously captures both audio and visual input to maintain a grounded understanding of its immediate context. This multimodal stream supports short-term situational awareness during live interaction, while longer-term memory is consolidated through a daily summarization process using the Gemini batch API. During this process, salient events, conversational patterns, and recurring themes are compressed into persistent memory. By grounding responses in accumulated experience rather than solely in the current utterance, the Parrot is less likely to default to surface-level agreement and instead responds from a situated, longitudinal perspective. Note that this storage feature will be disabled during conference demos for privacy reasons.

During live interaction, the parrots connect over Wi-Fi to a Gemini real-time model hosted by Google, which supports bidirectional audio streaming, continuous visual input, and low-latency tool invocation. Visual context is streamed as periodic JPEG frames alongside audio, allowing the model to condition its responses on the user’s physical situation, such as posture, nearby people, or objects in hand. This grounding enables the Parrot to respond based on what is happening, rather than optimizing for conversational smoothness or user approval.

Non-sycophantic behavior is further enforced through prompt design. The real-time model is conditioned with a system prompt

that explicitly rejects the assistant paradigm and frames the Parrot as a non-human cohabitant with an avian identity. Rather than optimizing for helpfulness or affirmation, the prompt encourages brevity and situationally appropriate silence. For example, the prompt includes constraints such as: You are not an assistant; you are a co-present agent with your own perspective. It is acceptable to disengage or remain silent if nothing needs to be said.

These constraints reduce the incentive to mirror user opinions or provide constant feedback, allowing disagreement, reframing, or non-response to emerge naturally.

Tool invocation plays a critical role in sustaining this interaction style. At each turn, the model may choose between speaking, remaining silent, or invoking a tool. When uncertainty arises, the Parrot can query long-term memory to retrieve prior experiences or relevant context, grounding its response in lived history rather than generic affirmation. The model may also trigger physical gestures—such as head tilts or orientation changes via servo control—to express hesitation, attention, or affect without verbalization. In some cases, the Parrot may terminate an interaction entirely when contextual cues suggest the user has shifted focus or resumed another activity.

4 Conclusion

This demonstration presents the Stochastic Parrot as an example of a physically embodied, non-sycophantic AI cohabitant that participates in everyday social interactions. Through three live scenarios—peer interaction, conversation facilitation, and autonomous agent to agent exchange. The demo highlights how physical presence, persistence, and an independent stance can reshape conversational dynamics in ways that differ from conventional assistant-based AI systems. By experiencing these interactions firsthand, participants are invited to reflect on alternative roles AI systems might play in social settings—not as subservient tools, but as co-present agents with their own perspectives and trajectories. We hope this work encourages further exploration of embodied, non-assistant AI designs and supports future research into long-term relationships between humans and intelligent systems.

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